GETTING STARTED IN R: WHAT IT DOES AND HOW IT DOES IT

ST331

(Statistical Computing 3)

INTRODUCTION TO R

- R is a computer language. It is a tool for doing computation.
- R studio is an open source integrated development environment (IDE) for creating and running R code. It's available for windows, Mac and Linux. Although you don't need an IDE to work with R , R studio makes life a lot easier.

DOWNLOADING R

- First things first. Download R from the Comprehensive R Archive Network (CRAN). In your browser, type this address if you work in Windows:

cran.r-project.org/bin/windows/base/

- Type this one if you work on the Mac:

cran.r-project.org/bin/macosx/

- Click the link to download R. This puts the win.exe file in your Windows computer, or the .pkg file in your Mac.

DOWNLOADING RSTUDIO

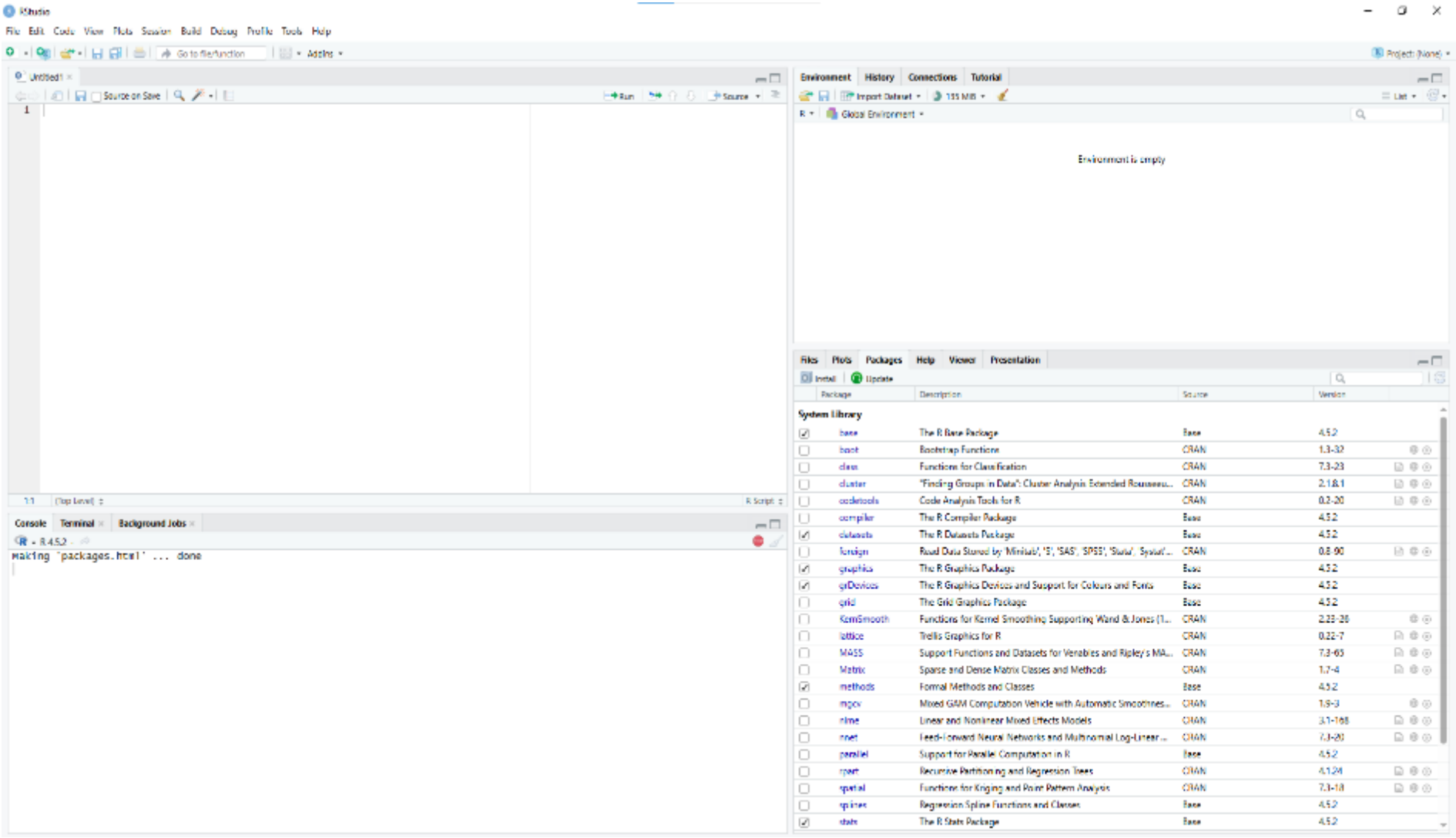
- For RStudio. Here's the URL:

  http://www.rstudio.com/products/rstudio/download

- Click the link for the installer for your computer, and again follow the usual installation procedures.

- After the RStudio installation is finished, click the RStudio icon to open the window shown below.

R STUDIO INTERFACE



R STUDIO INTERFACE EXPLAINED

- The Rstudio is an IDE for the R programming language that organizes its features into four panes.

- The source: For writing and editing codes
- The console: For running code interactively
- The environment/History pane: For tracking variables, functions and commands
- The files/plots/packages/Help pne: For managing files, viewing plots, viewing the packages and accessing help centers.

WORKING DIRECTORY

- What exactly does R save and where does R save it?
- In Windows, the default working directory is

C:\Users\<User Name>\Documents

- In Mac, the default working directory is

C:/Users/<User Name>/Documents

- You can also write a Windows file path in R as

C:\\Users\\<User Name>\\Documents

WORKING DIRECTORY

- If you ever forget the path to your working directory, type

```
getwd()
```

- in the Console pane, and R returns the path onscreen.
- If you like, you can change the working directory:

```
setwd(<filepath>)
```

ARITHMETIC, ASSIGNMENT & VECTORS

ARITHMETICS

- Here's an example in the console:

```
> 2+3
[1] 5
> 14/6
[1] 2.333333
> 14/6+5
[1] 7.333333
> 14/(6+5)
[1] 1.272727
> 3^2
[1] 9
> 2^3
[1] 8
```

- You can find the square root of any non-negative number with the sqrt function.

```
> sqrt(x=9)
[1] 3
> sqrt(x=5.311)
[1] 2.304561
> sqrt(9)
[1] 3
```

ARITHMETIC, ASSIGNMENT & VECTORS

ARITHMETICS

$$10^2 + \frac{3 \times 60}{8} - 3$$

$$\frac{5^3 \times (6 - 2)}{61 - 3 + 4}$$

$$2^{2+1} - 4 + 64^{-2^{2.25-\frac{1}{4}}}$$

$$\left(\frac{0.44 \times (1 - 0.44)}{34}\right)^{\frac{1}{2}}$$

ARITHMETIC, ASSIGNMENT & VECTORS

SOLUTION

```
> 10^2+3*60/8-3
[1] 119.5
> 5^3*(6-2)/(61-3+4)
[1] 8.064516
> 2^(2+1)-4+64^((-2)^(2.25-1/4))
[1] 16777220
> (0.44*(1-0.44)/34)^(1/2)
[1] 0.08512966
```

ARITHMETIC, ASSIGNMENT & VECTORS

Logarithms and Exponentials

- f (x) = e^x
- $log_e 20.08554$

```
> exp(x=3)
[1] 20.08554
> exp(3)
[1] 20.08554
> log(x=20.08554)
[1] 3
```

- $log_3 243$
- $log_e 20.08554$

```
> log(x=20.08554, base= 2.7182818)
[1] 3
> log(x=243,base=3)
[1] 5
```

CLASS EXERCISE

a. Using R, verify that

$$\frac{6a + 42}{3^{4.2-3.62}} = 29.50556$$

when $a = 2.3$.

b. Which of the following squares negative 4 and adds 2 to the result?

i. `(-4)^2+2`
ii. `-4^2+2`
iii. `(-4)^(2+2)`
iv. `-4^(2+2)`

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ARITHMETIC, ASSIGNMENT & VECTORS

Assigning Objects

```
> x <- -5
> x
[1] -5
> x = x + 1 # this overwrites the previous value of x
> x
[1] -4
> mynumber = 45.2
> y <- mynumber*x
> y
[1] -180.8
> ls()
[1] "mynumber" "x"          "y"
```

CLASS EXERCISE

a. Create an object that stores the value $3^2 \times 4^{1/8}$.

b. Overwrite your object in (a) by itself divided by 2.33. Print the result to the console.

c. Create a new object with the value -8.2×10^{-13}.

d. Print directly to the console the result of multiplying (b) by (c).

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